

D313

Anatomy and Physiology II with Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

In a simulated dissection, a student traces blood that has just returned from the body tissues through the venae cavae. The specimen heart weighed about 300 grams. Which chamber will pump this blood toward the lungs?

- A. The right ventricle
- B. The right atrium
- C. The left atrium
- D. The left ventricle

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which vessel carries oxygen-poor blood away from the heart?

- A. The pulmonary artery
- B. The aorta
- C. No vessel; all arteries carry oxygen-rich blood
- D. The pulmonary vein

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 45-year-old at rest has a heart rate of 72 beats per minute, a blood pressure of 118/76 mm Hg, and a stroke volume of 70 mL. What is her cardiac output?

- A. 8,260 mL/min
- B. 70 mL/min
- C. 5,040 mL/min (about 5.0 L/min)
- D. 142 mL/min

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A trained cyclist has a cardiac output of 6.0 L/min at a heart rate of 80 beats per minute. What is the stroke volume?

- A.** 480 mL
- B.** 75 mL
- C.** 80 mL
- D.** 6,000 mL

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

In the alveoli the partial pressure of oxygen is about 104 mm Hg, and in the arriving capillary blood it is about 40 mm Hg. The subject is a swimmer who trains five days a week. How does oxygen move into the blood?

- A.** Faster only because the swimmer trains often.
- B.** By simple diffusion from the alveoli into the blood.
- C.** From the blood into the alveoli.
- D.** By active transport using ATP pumps.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

During a relaxed breath, the diaphragm contracts and flattens. The student being observed is seated in a quiet lab room kept at 21 °C. What happens next?

- A.** Lung pressure rises above air pressure, so air flows in.
- B.** Air rushes in first, which then pushes the diaphragm down.
- C.** Chest volume increases, lung pressure drops below air pressure, and air flows in.
- D.** Chest volume decreases and air is pushed out.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

During a quiet exhalation, what causes air to leave the lungs?

- A.** The diaphragm contracts and pushes air out.
- B.** The diaphragm relaxes, chest volume falls, and lung pressure rises above air pressure.
- C.** Lung pressure drops below air pressure.
- D.** Abdominal muscles must always contract forcefully.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

After a long hike without water, a hiker is dehydrated. He is 28 years old and carried a 20-pound pack. His pituitary releases more ADH. What happens to his urine?

- A.** No change, because ADH acts only on blood vessels.
- B.** More sodium is secreted into the urine.
- C.** Larger volume and more dilute.
- D.** Smaller volume and more concentrated.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Alcohol suppresses ADH release. What is the expected effect on urine?

- A.** Smaller volume of concentrated urine.
- B.** More glucose in the urine.
- C.** No urine is produced.
- D.** Larger volume of dilute urine.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A student eats a cheeseburger for lunch at a campus café that opened last year. Which statement correctly describes how the fat in the meal is processed?

- A.** Bile is the enzyme that breaks fat into fatty acids.
- B.** Most fat is digested and absorbed in the stomach.
- C.** Fat is absorbed directly into the blood without being broken down.
- D.** Bile emulsifies the fat and pancreatic lipase digests it in the small intestine.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

After breast cancer surgery that removed several axillary lymph nodes, a patient develops swelling in her arm. She is left-handed and works as an accountant. Why does the swelling occur?

- A.** Fluid that leaks from capillaries cannot drain back through the damaged lymphatic pathway.
- B.** Heavy use of her dominant hand strained the arm.
- C.** The heart can no longer pump lymph to the arm.
- D.** Arteries in the arm were blocked.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

On day 14 of a 28-day cycle, a woman ovulates. She tracks her cycle with a phone app and swims twice a week. Which hormone event directly triggers ovulation?

- A.** A rise in FSH.
- B.** Oxytocin release.
- C.** A surge in LH from the anterior pituitary.
- D.** A rise in progesterone from the corpus luteum.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

In a simulated dissection, a student traces blood that has just returned from the body tissues through the venae cavae. ~~The specimen heart weighed about 300 grams.~~ Which chamber will pump this blood toward the lungs?

Target word: chamber that pumps to lungs **Decides:** Blood from the venae cavae enters the right side.

A	Correct	Right atrium → right ventricle → pulmonary trunk to the lungs.
B	Dropped element	The atrium receives the blood but the ventricle pumps it to the lungs.
C	Confusion set	The left atrium receives blood returning from the lungs.
D	Inverted operation	The left ventricle pumps oxygen-rich blood to the body.

2. Answer A Concept 1 • transfer

Which vessel carries oxygen-poor blood away from the heart?

Target word: reverse: find the exception **Decides:** Arteries carry blood away; the pulmonary artery carries oxygen-poor blood.

A	Correct	It carries oxygen-poor blood from the right ventricle to the lungs.
B	Adjacent category	Carries oxygen-rich blood away from the heart.
C	Absolute language	The pulmonary arteries are the exception.
D	Inverted operation	Carries oxygen-rich blood toward the heart.

3. Answer C Concept 2

~~A 45-year-old~~ at rest has a heart rate of 72 beats per minute, ~~a blood pressure of 118/76 mm Hg,~~ and a stroke volume of 70 mL. What is her cardiac output?

Target word: cardiac output **Decides:** $HR \times SV$.

A	Wrong stem number	Uses the systolic pressure 118×70 .
B	Dropped element	Stroke volume alone, without heart rate.
C	Correct	$72 \times 70 = 5,040 \text{ mL/min}$.
D	Inverted operation	Adds $72 + 70$ instead of multiplying.

4. Answer B Concept 2 • transfer

A trained cyclist has a cardiac output of 6.0 L/min at a heart rate of 80 beats per minute. What is the stroke volume?

Target word: solve for stroke volume **Decides:** $SV = CO / HR$.

A	Inverted operation	Multiplies 6.0×80 instead of dividing.
B	Correct	$6,000 \text{ mL} / 80 = 75 \text{ mL}$.
C	Wrong stem number	The heart rate, not the stroke volume.
D	Dropped element	The cardiac output, not divided by heart rate.

5. Answer B Concept 3

In the alveoli the partial pressure of oxygen is about 104 mm Hg, and in the arriving capillary blood it is about 40 mm Hg. ~~The subject is a swimmer who trains five days a week.~~ How does oxygen move into the blood?

Target word: mechanism of gas exchange **Decides:** O₂ moves from higher to lower partial pressure.

A	Camouflage	Training does not change the diffusion mechanism.
B	Correct	O ₂ moves down its partial pressure gradient; no energy needed.
C	Inverted operation	That is the direction for CO ₂ , not O ₂ .
D	Adjacent category	Gas exchange is passive.

6. Answer C Concept 4

During a relaxed breath, the diaphragm contracts and flattens. ~~The student being observed is seated in a quiet lab room kept at 21 °C.~~ What happens next?

Target word: inspiration sequence **Decides:** Volume up → pressure down → air flows in.

A	Dropped element	Drops the inverse link between volume and pressure.
B	Confusion set	Pressure change causes airflow, not the reverse.
C	Correct	Boyle's law: volume and pressure change inversely.
D	Inverted operation	That describes exhalation when the diaphragm relaxes.

7. Answer B Concept 4 • transfer

During a quiet exhalation, what causes air to leave the lungs?

Target word: reverse: expiration **Decides:** Diaphragm relaxes, volume falls, pressure rises.

A	Inverted operation	Contraction enlarges the chest and draws air in.
B	Correct	Smaller volume means higher pressure, so air flows out passively.
C	Confusion set	That drives inhalation.
D	Absolute language	Quiet exhalation is passive; forced exhalation uses those muscles.

8. Answer D Concept 5

After a long hike without water, a hiker is dehydrated. ~~He is 28 years old and carried a 20-pound pack.~~ His pituitary releases more ADH. What happens to his urine?

Target word: effect of high ADH **Decides:** ADH increases water reabsorption.

A	Adjacent category	ADH's main kidney target is the collecting duct.
B	Confusion set	Sodium handling is mainly aldosterone and ANP.
C	Inverted operation	That is the effect of low ADH.
D	Correct	More water is reabsorbed in the collecting ducts.

9. Answer D Concept 5 • transfer

Alcohol suppresses ADH release. What is the expected effect on urine?

Target word: reverse: low ADH **Decides:** Less water reabsorbed.

A	Inverted operation	That is the effect of high ADH.
B	Confusion set	Glucose in urine relates to blood sugar and reabsorption limits.
C	Absolute language	Filtration continues; output rises.
D	Correct	Without ADH, collecting ducts reabsorb less water.

10. Answer D Concept 6

A student eats a cheeseburger for lunch ~~at a campus café that opened last year~~. Which statement correctly describes how the fat in the meal is processed?

Target word: fat digestion **Decides:** Bile emulsifies; lipase digests in the small intestine.

A	Confusion set	Bile is an emulsifier, not an enzyme.
B	Adjacent category	The stomach starts protein digestion; fat is handled in the small intestine.
C	Dropped element	Skips emulsification and lipase digestion.
D	Correct	Bile increases surface area; lipase is the enzyme.

11. Answer A Concept 7

After breast cancer surgery that removed several axillary lymph nodes, a patient develops swelling in her arm. ~~She is left handed and works as an accountant~~. Why does the swelling occur?

Target word: lymphatic function **Decides:** Lymph vessels return interstitial fluid to the blood.

A	Correct	Lymphatics return excess tissue fluid; disruption causes lymphedema.
B	Camouflage	Handedness does not explain lymphedema.
C	Confusion set	Lymph is not pumped by the heart.
D	Adjacent category	Arterial blockage causes ischemia, not this kind of swelling.

12. Answer C Concept 8

On day 14 of a 28-day cycle, a woman ovulates. ~~She tracks her cycle with a phone app and swims twice a week~~. Which hormone event directly triggers ovulation?

Target word: ovulation trigger **Decides:** The LH surge.

A	Confusion set	FSH drives follicle growth, not rupture.
B	Adjacent category	Oxytocin drives labor and milk letdown.
C	Correct	The mid-cycle LH surge causes the follicle to rupture.
D	Inverted operation	The corpus luteum forms after ovulation.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	2	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Path of blood through the heart	Right atrium → tricuspid → right ventricle → pulmonary valve → pulmonary arteries → lungs → pulmonary veins → left atrium → mitral → left ventricle → aortic valve → aorta.	Assuming arteries always carry oxygen-rich blood.	A question about which chamber or vessel handles oxygen-poor or oxygen-rich blood.
2	Cardiac output	Cardiac output = heart rate × stroke volume.	Adding the values or plugging in blood pressure.	A heart rate in beats per minute and a stroke volume in mL.
3	Gas exchange by diffusion	O ₂ and CO ₂ move by simple diffusion from higher to lower partial pressure across the alveolar and capillary walls.	Thinking gases are pumped actively.	Partial pressures on either side of the respiratory membrane.
4	Mechanics of breathing	When the diaphragm contracts, chest volume rises and pressure in the lungs falls below air pressure, so air flows in (Boyle's law); relaxation reverses this.	Thinking air rushing in causes the chest to expand.	Diaphragm action paired with pressure or volume changes.
5	ADH and urine concentration	ADH makes the collecting ducts more permeable to water, so more water is reabsorbed and urine becomes concentrated; low ADH produces large volumes of dilute urine.	Thinking ADH increases urine output.	Dehydration, alcohol, or ADH level and urine volume.
6	Digestion and absorption sites	Protein digestion begins in the stomach (pepsin); most chemical digestion and nutrient absorption happen in the small intestine; bile emulsifies fat but is not an enzyme.	Calling bile a fat-digesting enzyme or the stomach the main absorption site.	Where a nutrient is broken down or absorbed.
7	Lymphatic system roles	Lymphatic vessels return excess interstitial fluid to the blood, absorb dietary fat (lacteals), and carry lymph through nodes that filter it for immune cells.	Thinking lymph circulates in a closed loop pumped by the heart.	Swelling after node removal or a question about fluid return.
8	Reproductive hormone control	FSH stimulates follicle growth; a mid-cycle LH surge triggers ovulation; testosterone and estrogen feed back to regulate GnRH, FSH, and LH.	Crediting FSH for ovulation.	A specific cycle event and which hormone causes it.

Course scope (public description): This is Anatomy and Physiology II, a six section, four CEU course that enables students to develop an understanding of the relationships between the structures and functions of the cardiovascular, respiratory, digestive, urinary, reproductive, and lymphatic systems in the human body. This course will involve laboratory activities, simulated dissections, textbook material, models, and diagrams. Because the course is self-paced, you may move through the material as quickly or as slowly as you need to, with the goal

of demonstrating proficiency in the four competencies covered in the final assessment. If you have no prior knowledge of this material, you can expect to spend 40–60 hours on the course content.